

Prenatal Syphilis

Neonatal Period

- **Aqueous penicillin G:** 50,000 units/kg intravenously
- Every 12 hours for the first 7 days of life
- Every 8 hours for the next 3 days
- **Procaine penicillin G:** 50,000 units/kg intramuscularly daily for 10 to 14 days

Postneonatal Period

- If **CSF examination is negative:**
- **Benzathine penicillin G,** 50,000 units/kg (up to 2.4 million units) intramuscularly
- If **CSF examination is positive:**
- **Aqueous penicillin G,** 50,000 units/kg every 4 to 6 hours intravenously for 10 to 14 days

Presumed Congenital Syphilis

Infants should be treated for presumed congenital syphilis regardless of evaluation if born to mothers with any of the following: - Untreated syphilis at delivery

- Treatment other than with penicillin
- Treatment for syphilis within 1 month of delivery
- Inadequate documentation of appropriate treatment
- Serologic evidence of relapse or reinfection after treatment
- No documented fourfold decrease in RPR titers after treatment before or during pregnancy

Evaluation of the Infant

If the infant has: - Signs of congenital syphilis on physical examination, or
- Serologic evidence of syphilis (infant's RPR titer $\geq$ two dilutions higher than mother's)

Then perform: - Complete blood count with differential and platelets
- CSF examination (cell count, protein, RPR)

Additional tests as clinically indicated: - Long-bone radiographs

- Chest radiograph
- Cranial ultrasound
- Ophthalmologic examination
- Auditory-brainstem response
- Liver function tests

Management Based on Evaluation

- If infant's clinical, CSF, serologic, and blood evaluations are normal **and** maternal treatment compliance is confirmed:
- Administer a single intramuscular injection of **benzathine penicillin G**, 50,000 units/kg
- If mother was adequately treated with appropriate serologic decline and infant's nontreponemal titer is:
- **Equal to or less than mother's titer:** No further evaluation needed; give single dose of **benzathine penicillin G**
- **Nonreactive:** No treatment necessary

Follow-Up

All seropositive infants or infants born to seroreactive mothers at delivery should undergo: - Clinical and serologic re-evaluation every 2–3 months
- Until nontreponemal test becomes nonreactive or titer decreases fourfold

Expected Serologic Response

- If infection is due to **passive transplacental antibody transfer**:
- Titers should decline by 3 months and become nonreactive by 6 months
- If infant had **true congenital syphilis (infected and treated)**:
- Serologic decline may be slower

Indications for Re-evaluation and Retreatment

- If titers remain **stable or increase after 8–12 months of age**:
- Re-evaluate (including CSF analysis)
- Retreat with 10-day course of intramuscular penicillin G

Treponemal Test Interpretation

- May remain positive up to **15 months** after transplacental acquisition
- If still reactive **after 18 months**, indicates congenital syphilis

Neurologic Follow-Up

For infants with initially abnormal CSF: - Perform serial lumbar punctures every 6 months until CSF cell count normalizes

- If CSF pleocytosis persists beyond **2 years** or no downward trend is observed:
- Retreat
- Check CSF VDRL at 6 months
- If still reactive, administer retreatment

Treatment of Sex Partners and Persons Exposed to Syphilis

Evaluation

- Examine exposed person
- Perform serologic testing

Treatment Indications

Treat contacts who have: - Clinical or serologic evidence of syphilis

Prophylactic Treatment for Primary Syphilis

Indicated if: - Exposure occurred **within 90 days** before partner's diagnosis of early syphilis (primary, secondary, or early latent)

- Note: Patients without clinical disease but with serologic titers $\geq 1:32$ are considered to have early syphilis
- Exposure occurred **>90 days** before partner's diagnosis of early syphilis, but:
- Serologic tests are not immediately available
- Test reliability is uncertain
- Long-term sex partner of a person with latent syphilis and evaluation shows possible serologic or clinical transmission